

Technical Portfolio

Internship Projects and Personal Studies

This portfolio presents a selection of academic, industrial, and personal projects, carried out as part of internships and self-directed studies, with a particular focus on numerical simulation, thermomechanical analysis, and mechanical design.

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1 General Electric Vernova

1.1 Project 1 : Stiffness Measurement Station

1.1.1 Objective

Design (mechanical and software components) of a stiffness measurement station for the finger contact springs of the new series of live tank circuit breakers. The purpose of this station is to ensure the conformity of the finger contact springs after assembly in the housing.

1.1.2 Execution

My responsibility was to develop the station in its entirety, both mechanically and in terms of software. I began by defining requirements with the various stakeholders (quality department, operators, design office). These needs were translated into system-level requirements, then broken down into subsystem-level requirements and further into component-level requirements (top-down design approach).

Once the component-level requirements were defined, I began the design phase while simultaneously working with my supervisor to identify the simulations needed to validate the measurement accuracy requirements. After dimensional validation, I produced the engineering drawings, which were then sent to the procurement department.

The software component was developed in parallel. I created a graphical user interface for data visualization and control of the force sensor, then synchronized it with the site database for data logging.

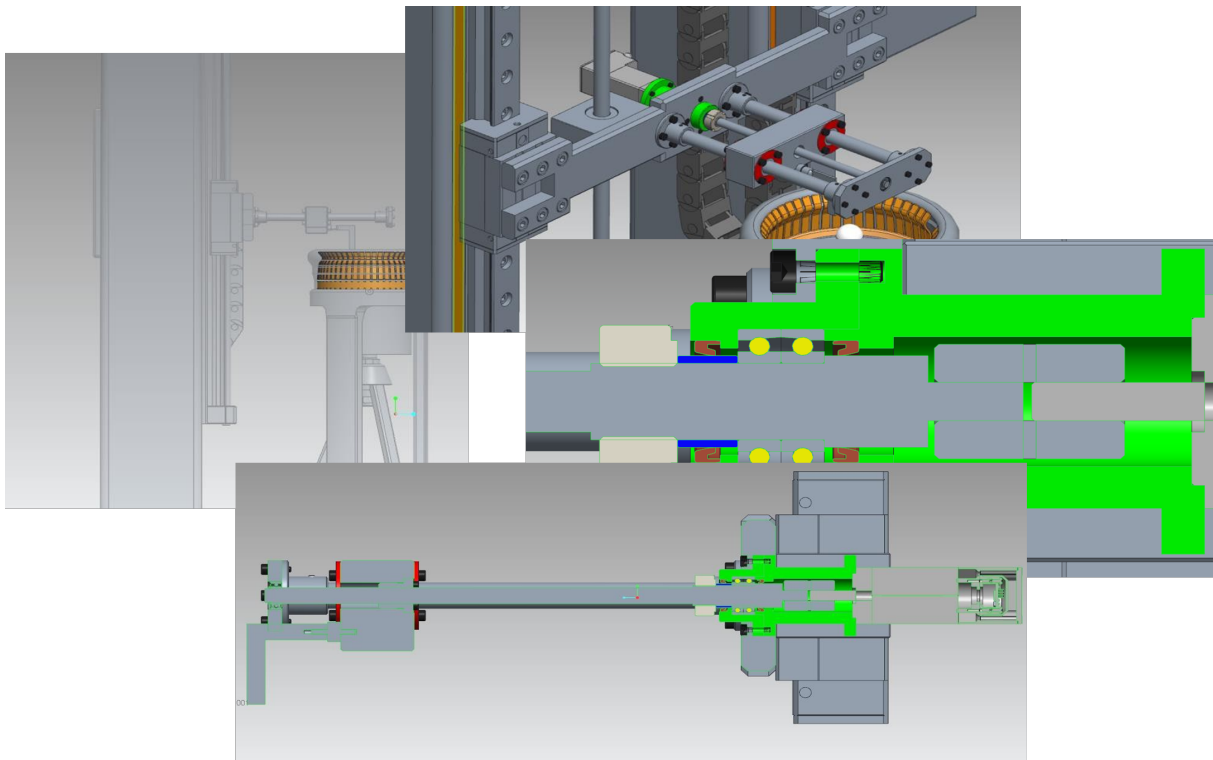


FIGURE 1 – Mechanical design of the measurement station

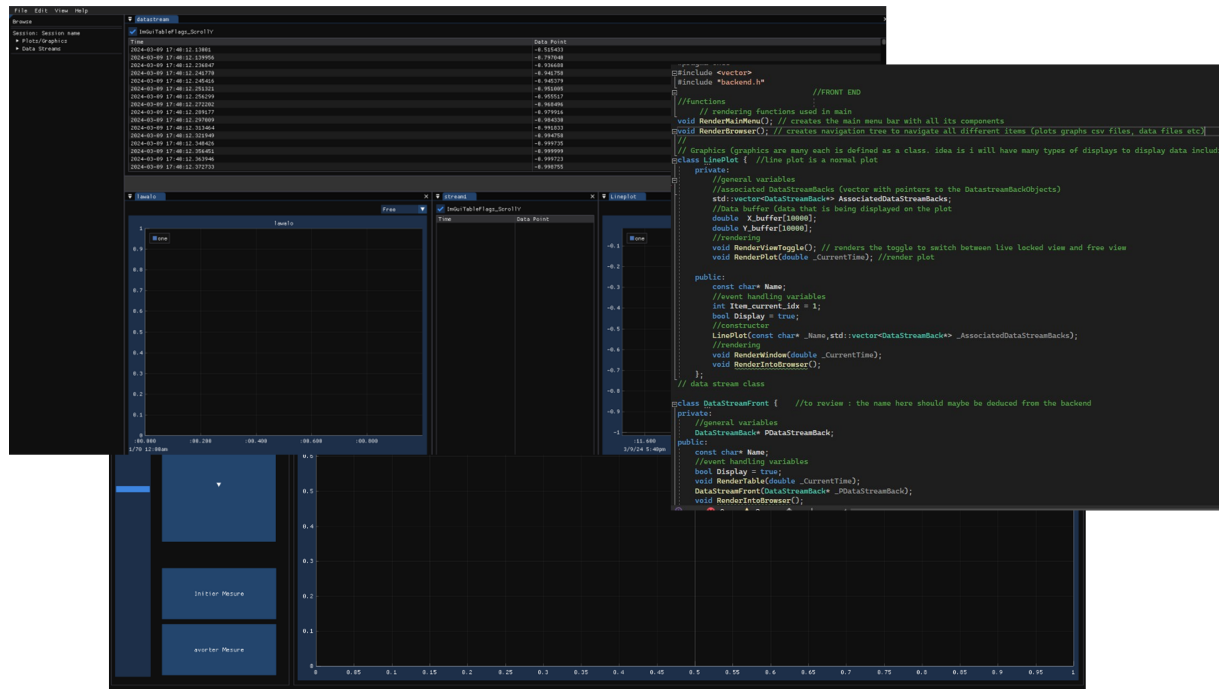


FIGURE 2 – Software interface and data acquisition system

1.2 Project 2 : Power Transmission Shaft Sizing

1.2.1 Objective

Sizing of a redesigned power transmission shaft for the opening mechanism of GCB (Ground Circuit Breaker) circuit breakers. The previous system was costly and sensitive to coaxiality defects.

1.2.2 Execution

I sized a new shaft design by first simulating the various proposals made by a colleague. I iterated over different dimensions and made minor modifications to achieve the required safety factor without exceeding it. The redesign aimed to simplify machining operations and, most importantly, to reduce the amount of material used, which represented a significant cost for the company.

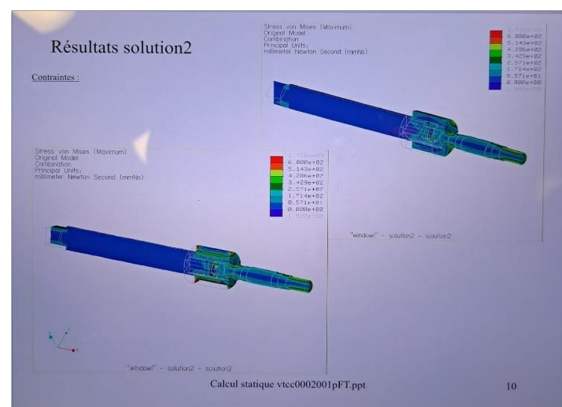
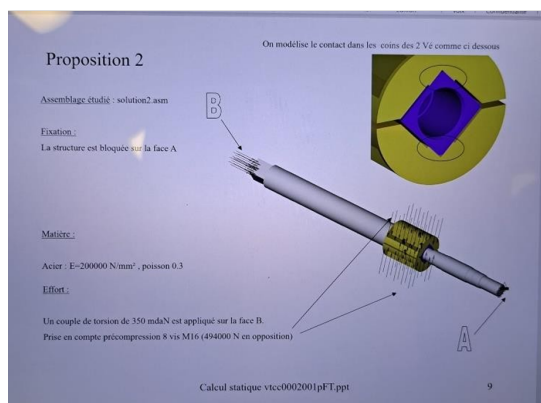
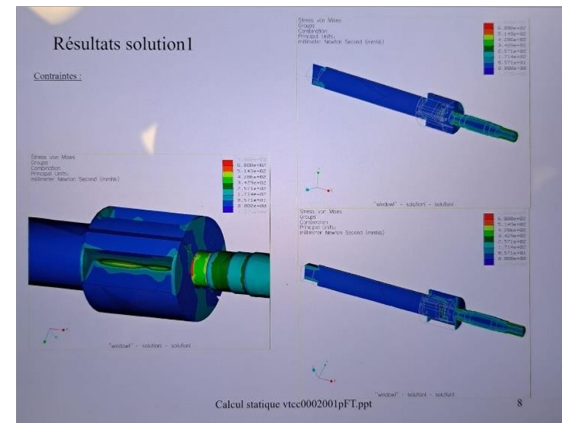
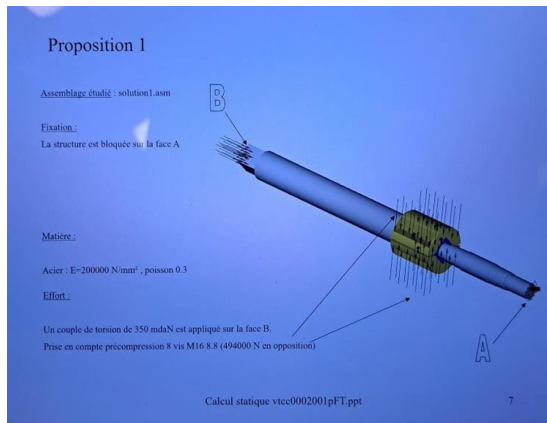


FIGURE 3 – FEA analysis of the initial design

1.3 Project 3 : Stress Analysis of SF6 Gas Chamber

1.3.1 Objective

Determine the mechanical stress in the SF6 gas chamber due to the gas pressure, which was increased in order to extend the service life of the tulip contacts. The objective is to ensure that the safety factor remains above 5 as specified in the design requirements (CDC 912625/CODAP).

1.3.2 Execution

The load was applied to the inner walls to simulate a static pressure of $p = 1.6$ MPa. Material used : A-S7G03.

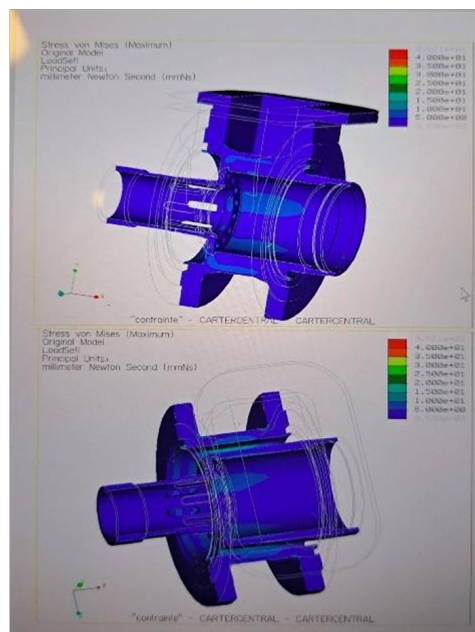
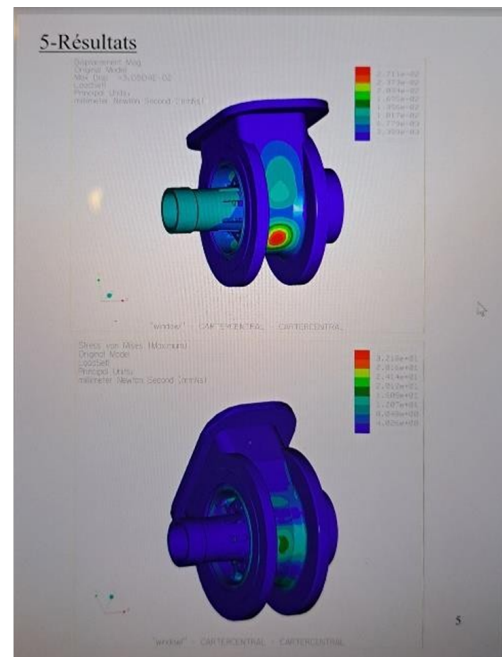
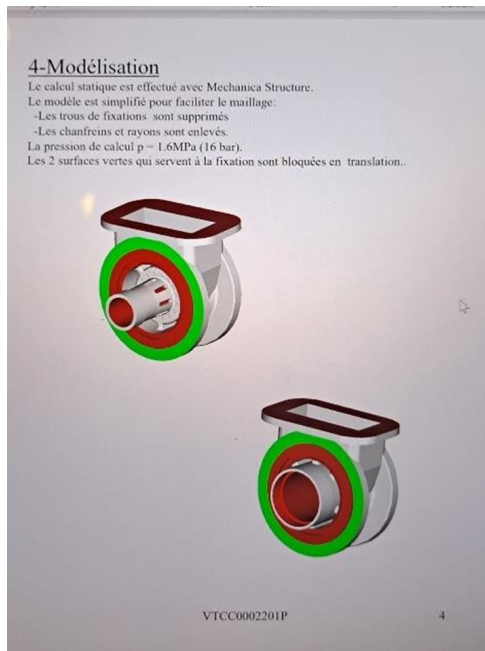


FIGURE 4 – FEA model of the gas chamber

2 Caterpillar : Final Year Internship

2.1 Objective and Context

The Caterpillar D5 bulldozer had been accumulating field failures on its chain tensioner assembly, grease leaks, cracked plugs, worn retainer plates, premature piston-seal failures. The product-support team had flagged the problem but the failure modes had never been quantified, and the dominant root causes were still unclear. My internship mission was to close that loop : diagnose the system end-to-end, validate the diagnosis with simulation, and deliver redesign concepts that addressed the root causes while remaining compatible with manufacturing and assembly constraints.

The project followed a full **DMAIC cycle**. Scope was restricted to the D5 variants (D5 LGP, D5 XL) and specifically to the tensioner group . The D4, D6, D7, John Deere 700L and Komatsu D51 PX systems were used as benchmark references. Deliverables were 3D CAD redesigns validated in FEA, not physical prototypes. Tools : Creo Parametric for CAD, ANSYS Mechanical for structural FEA, Python for data processing.

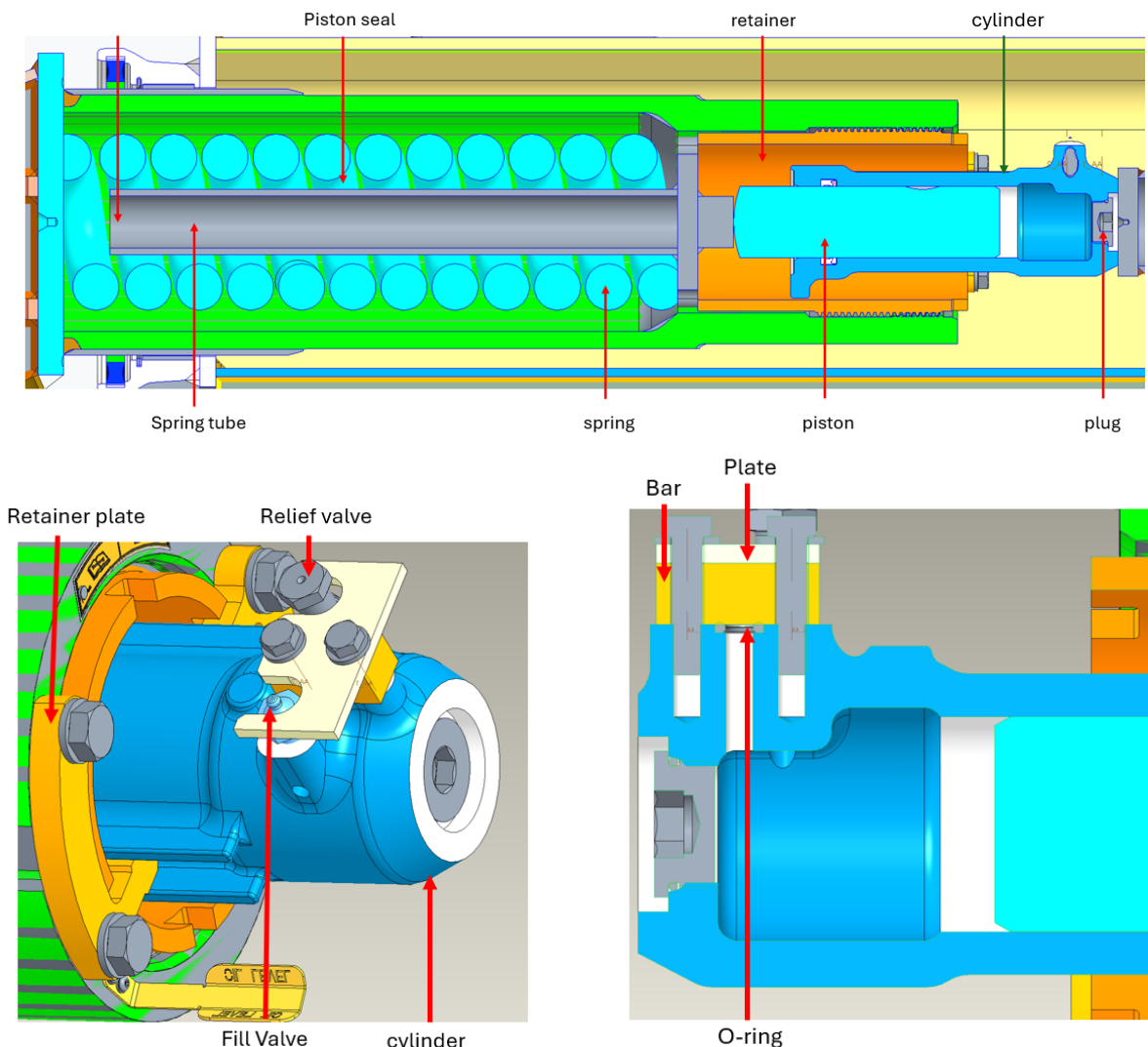


FIGURE 5 – D5 track tensioner : cross-section showing the main components involved in the failure modes (spring tube, piston, piston seal, retainer, cylinder, plug). Grease pressure acts on the piston base to tension the track ; the plug seals the rear of the cylinder and is bolted to the housing.

2.2 Automated Failure-Mode Classification of Warranty Reports

Caterpillar maintains a centralised warranty database : every time a dealer repairs a machine under warranty, they log the failed part, the machine hours, a short *comment* field and a longer *claim story*. Filtering by the tensioner part numbers returned records **in the order of thousands** a dataset large enough to make statistical analysis meaningful, but far too large to read manually. The real obstacle was that dealers worldwide write in their native language : English, German, Polish, Japanese, Spanish, sometimes with abbreviations and regional shorthand. Per-component *frequency* was easy to compute from the part number alone, but extracting the actual *failure mode* from free text required something smarter.

My first attempt used the Gemini API : feed each concatenated `comment + claim_story` to the model with a system prompt asking it to pick one mode from a fixed list. It worked well, but every call sent customer-written warranty text to a third-party server. Even though the raw text did not mention "Caterpillar" or part numbers, sending it out over an API was not acceptable for this use case. Deploying a local LLM server (Ollama-style) would have solved the privacy issue but required IT approvals I was not going to get in time.

I pivoted to a lighter, fully local approach based on **semantic similarity with Sentence-BERT**. SBERT embeds a sentence into a dense high-dimensional vector whose geometry captures semantic meaning two sentences expressing the same idea in different words (or different languages) land close together. I embedded each of the eight candidate failure modes (*grease leak, broken threads, detached part, loose part, broken/cracked part, corroded part, clogged, damaged threads/grooves*) once, then embedded every claim story, and classified each one by the argmax of cosine similarity :

$$\text{FM}(t) = \arg \max_i \frac{\mathbf{v}_t \cdot \mathbf{v}_{\text{MD}_i}}{\|\mathbf{v}_t\| \|\mathbf{v}_{\text{MD}_i}\|}$$

A similarity threshold was applied below which the case was marked null, so borderline or uninformative claim stories would not pollute the statistics. The `all-MiniLM-L6-v2` model runs comfortably on CPU and handled the multilingual content well enough that manual translation was unnecessary. The entire pipeline ran on my workstation, no data left the machine. I validated the output against 50 manually-labelled samples and the classifier hit **48/50 = 96%** accuracy, which was well within acceptable bounds for prioritisation work.

Crossing the resulting failure-mode distribution with each component's warranty cost revealed a clear ranking of critical failures : leakage at the piston/seal interface, leakage at the fill-valve/cylinder interface, plug loosening and cracking, and leakage at the relief-valve interface. Equally striking was what the data *did not* contain : zero reports of the intentional fuse system actuating, despite multiple plug-cracking cases. That contradiction became the focus of the next phase.

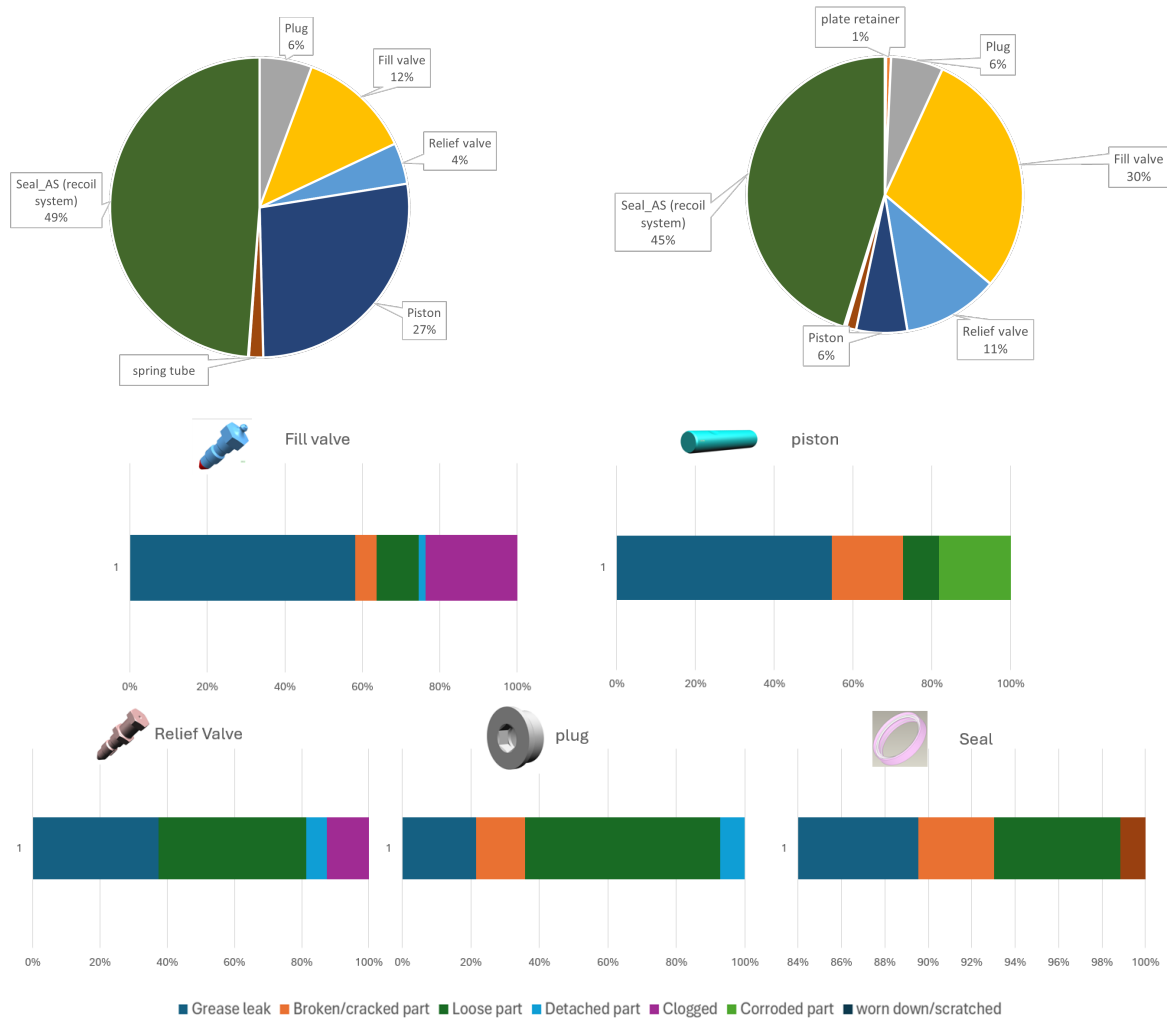


FIGURE 6 – Failure-mode distribution per component after automated classification of the warranty records. Grease leak dominates across seals, valves and plug.

2.3 FEA of the Fuse System : Why Plugs Were Cracking

The tensioner is built with an intentional mechanical fuse, a deformable steel bar backed by an O-ring designed to open a leak path *before* any other component reaches its yield limit. If the fuse was doing its job, cracked plugs should not exist in the warranty record. They did. Something in the actual stress response of the assembly was violating the intent of the design, and I needed an FEA model to find out what.

I built a coupled model of the fuse assembly in ANSYS Mechanical, applying grease pressure directly to the internal surfaces and pre-tensioning the bolts and plug per the design drawings. The first version modelled the O-ring explicitly as a hyperelastic body the theoretically correct choice. In practice the O-ring deformation became so large that mesh elements distorted past ANSYS's convergence tolerance, and the simulation crashed above 24 pressure units, far below the regime of interest (130–250 units).

Rather than fight the solver with finer meshes and remeshing hacks, I reformulated the leak criterion *geometrically*. Per the Parker O-ring Handbook, an elastomeric seal is guaranteed to hold as long as its compression exceeds 5.7%, and is guaranteed to leak below 0% (loss of contact). If I remove the O-ring from the simulation entirely and track the local vertical displacement $d_{z,k}$ of each node along the seal contour, I can compute a per-node compression :

$$C_{\%,k} = \frac{t_o - t_{c,k}}{t_o}, \quad t_{c,k} = d_{z,k} + g_d$$

where t_o is the free O-ring thickness and g_d is the gland depth. Aggregating over all contour nodes gives **guaranteed sealing** when $\min_k C_{\%,k} > 5.7\%$ and **certain leakage** when $\overline{C_{\%}} < 0\%$. With the O-ring removed, the model meshed cleanly and ran stable all the way to plastic yield, while still producing a physically meaningful leak prediction. The model also exploited the cylindrical symmetry of the assembly, halving the node count and the solve time.

The results were unambiguous. At roughly 130 pressure units right around the maximum service pressure recorded during bench testing the plug threads reached 329 MPa, already above the 310 MPa yield limit of the material. At that same pressure the fuse plate had barely deformed, and the O-ring compression was still sitting around 20% everywhere along the contour. Even pushing the simulation all the way to 240 units, the fuse still would not open a leak path. **The fuse was massively oversized and effectively inactive**, leaving the plug as the *de facto* weakest link exactly consistent with the warranty record.

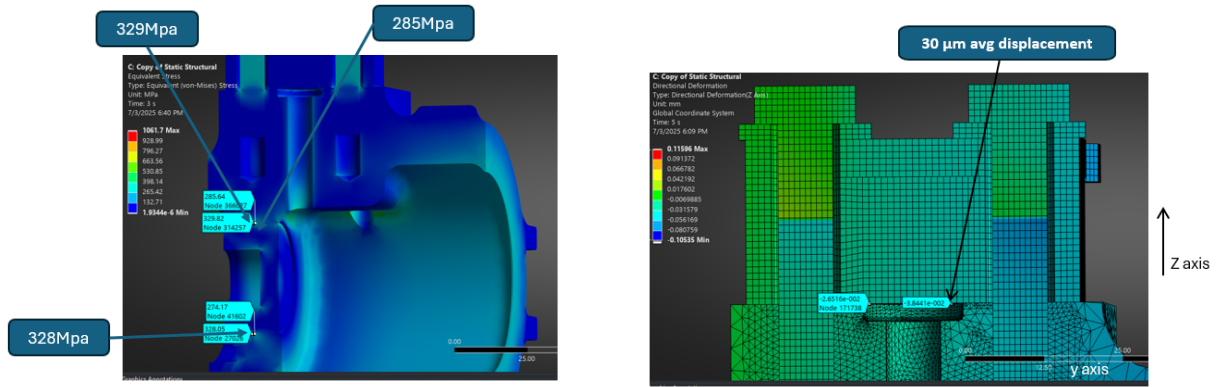


FIGURE 7 – Von Mises stress in the fuse assembly at 130 pressure units. The plug threads reach 329 MPa (yield = 310 MPa) while the fuse plate remains elastic, the fuse never triggers before plug failure.

2.4 Fuse Re-sizing

With the fuse proven ineffective, the cheapest corrective action was to keep the same topology but thin down the plate until it actually deformed in the right pressure window. The desired window had clear bounds : seal reliably above the 130-unit maximum service pressure, and leak reliably below the 220-unit cylinder yield pressure a 90-unit band to work inside. (The plug itself is weaker than the cylinder, but its redesign was out of scope for this sizing study ; once the plug is redesigned to match the cylinder, the same curves still apply.)

I re-ran the simulation parametrically across four supplier-standard plate thicknesses 3, 3.5, 4 and 6 mm and for each one extracted the minimum and average O-ring compression along the contact contour as pressure ramped. As expected, thinner plates leak earlier and produce wider grey-zone bands. The 6 mm plate essentially replicated the current oversized behaviour ; the 3 mm plate leaked too early, barely clearing the 130-unit service ceiling.

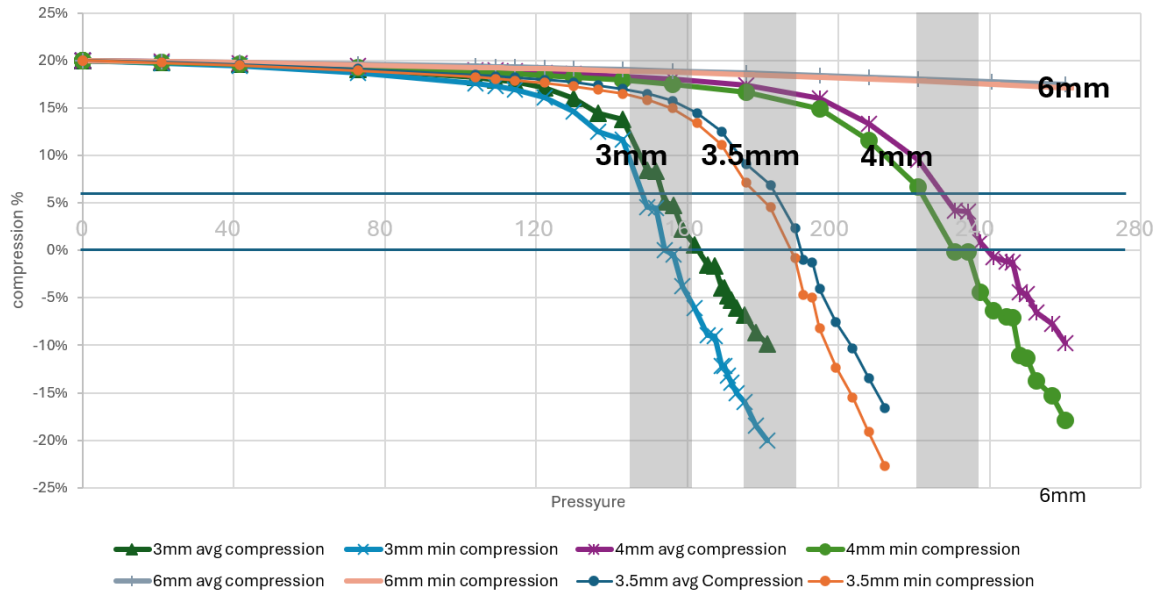


FIGURE 8 – O-ring compression vs. pressure for each plate thickness. Grey bands mark the transition between guaranteed sealing ($\min C\% > 5.7\%$) and certain leakage ($C\% < 0\%$).

The **3.5 mm plate** was the clean answer : guaranteed sealing up to around 160 units (comfortably above 130) and guaranteed leakage by around 180 units (comfortably below 220). Part cost stayed negligible since the geometry is still a simple stamped plate. One important limitation worth flagging : this fuse is designed for static or quasi-static overpressure. A sudden pressure spike, for example a hard idler impact would require a much higher mass-flow evacuation path than the small opening produced by plate deformation. For that failure mode a proper relief valve is the right tool, and that informed the concept-level redesign.

2.5 Redesign Concepts and Pugh-Matrix Selection

Three full CAD redesigns were developed in Creo Parametric, each informed by both the warranty analysis and a benchmark of the John Deere 700L, Komatsu D51 PX, and CAT D4/D7/TTL tensioners. The recurring themes across successful competitor designs were consistent : welded (not threaded) cylinder assembly to eliminate the fill-valve leak path, chromed or sleeved piston-cylinder interface to prevent wear-induced seal failure, oil-based lubrication of the sliding contact, a positive mechanical alignment between cylinder and frame, and a proper relief valve as the overpressure safety device rather than a deformable bar.

Concept 1 focused on a perfect-guidance architecture : spherical joint at the piston end, a spacer-and-sleeve retainer providing two line contacts against a precision-machined cylinder, and an interconnected oil chamber lubricating both interfaces. Concept 2 was inspired directly by the John Deere 700L : a two-piece welded cylinder, a re-sized fuse plate from the sizing study, and a guide hole catching the piston if the chain goes slack. Concept 3 took a different route , it integrates the cylinder body directly into the mobile portion of the TRF, uses a two-level alignment (ball joint at one end, pin-in-hole at the other) that makes the sub-assembly essentially self-aligning under its own weight during installation, and keeps both valves accessible without modifying the TRF hatch door.

The three concepts were scored against eight criteria in a Pugh matrix using the current D5 as the reference : cost, intrusiveness on the existing design, serviceability, operator safety, ease of assembly, behaviour under chain-loose scenarios, fuse robustness, and manufacturability. The matrix was intentionally kept qualitative , weighting the criteria with specific numerical coefficients would have introduced arbitrary bias given that several criteria (assembly ease especially)

were the blocking constraints.

Concept 3 was selected. It scored positively on cost, serviceability, manufacturability, intrusiveness and chain-loose guidance ; neutral on assembly ease, where its self-aligning geometry directly addresses what had been the single biggest pain point. Concept 2 was blocked by severe assembly difficulty inside the cramped TRF, and Concept 1 offered no clear advantage over the reference. Concept 3 is now positioned for physical prototyping and bench validation the remaining step in the DMAIC Control phase.

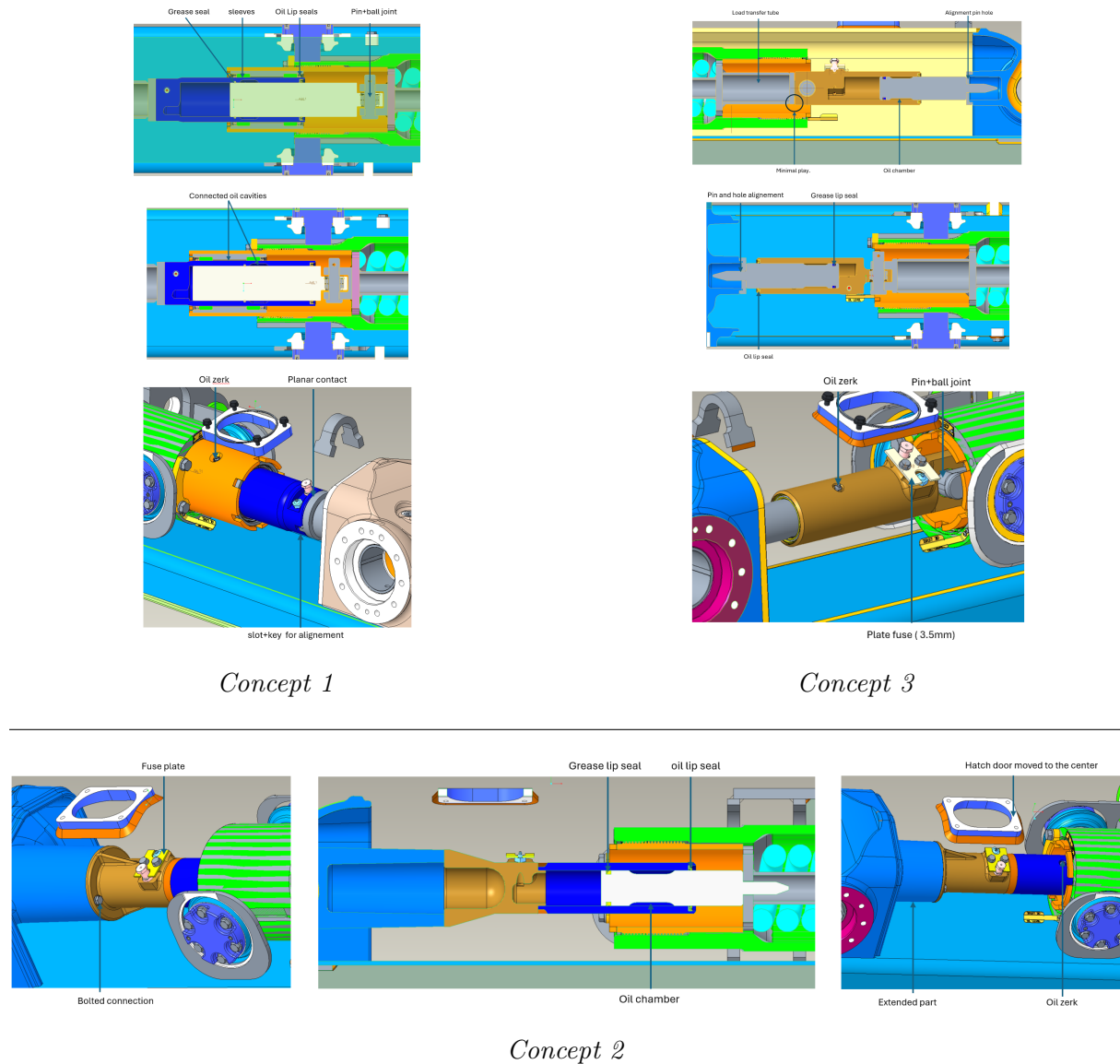


FIGURE 9 – Comparison of the three design concepts.

Key Outcomes

- 684 multilingual warranty reports classified automatically at **96% accuracy** with a local SBERT pipeline (no data leaves the machine).
- FEA with a geometric leak criterion proved the existing fuse never activates, and identified the plug as the real failure point matching field data exactly.
- Re-sized fuse plate (3.5 mm) delivers a clean pressure window : seals up to ~ 160 , leaks by ~ 180 , well inside the 130–220 safety band.
- Concept 3 selected via Pugh matrix, addressing all four critical failure modes identified upstream and ready for prototyping.

3 Personal Studies

Note

These are personal studies aimed at improving my skills in numerical simulation and thermomechanical analysis.

3.1 Study 1 : Optimization of Regenerative Cooling in a Combustion Chamber

3.1.1 Context

During operation, rocket engines particularly the combustion chamber are subjected to extremely high temperatures. As temperature increases, the mechanical properties of metals degrade, most notably their yield strength. To address this issue, two methods exist for controlling the temperature of the chamber :

- **Ablative cooling** : involves the gradual erosion of the inner chamber walls, carrying away heat in the process. However, this method is destructive and assumes the engine will not be reused.
- **Regenerative cooling** : the most commonly used method. It works by circulating a cryogenic fluid through channels machined into the combustion chamber wall, enabling efficient heat dissipation.

3.1.2 Study Objective

Evaluate the effect of varying the width L and height H of straight cooling channels on the maximum temperature of the inner combustion chamber wall. The objective is to find an optimum with the maximum inner wall temperature as the objective function and L and H as optimization variables, with :

- L between 0.8 and 3.0 mm
- H between 1.5 and 5.0 mm
- Wall thickness kept constant at 10 mm
- Distance between the channel and the inner wall kept constant at 3 mm

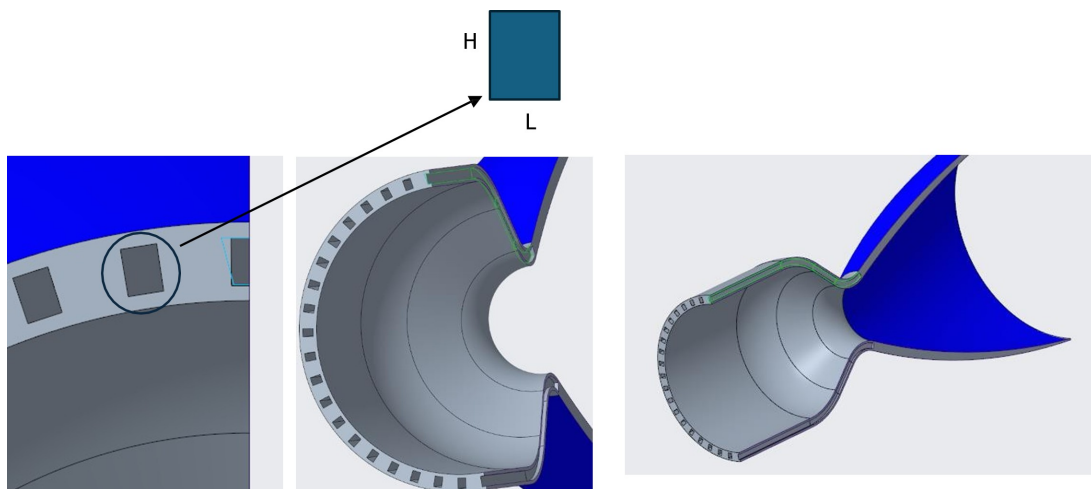


FIGURE 10 – Cooling channel design (CAD assembly of combustion chamber + nozzle)

3.1.3 Study Parameters

Parameter	Value
Fuel/oxidizer	Methane/liquid oxygen
Inner wall material	CuAgZr
O ₂ /fuel mixture ratio	3.6
Chamber pressure (MPa)	3
Coolant	N ₂ (liquid nitrogen)
Coolant mass flow rate (kg/s)	1.5
Coolant inlet temperature (K)	70
Contraction/expansion ratio	9/5.5
Wall thickness (mm)	10
Channel-to-wall margin thickness (mm)	3
Throat diameter (m)	0.0274

TABLE 1 – Simulation parameters

3.1.4 Execution

Initially, the simulation was carried out with an inner wall thickness, cooling channel width, and height of 1 mm, 1.8 mm, and 3.5 mm, respectively.

The mesh was generated with inflation layers at the boundary layer of the combustion chamber to capture temperature and velocity gradients, and similarly for the cooling channels. A mesh of the solid region was also generated.

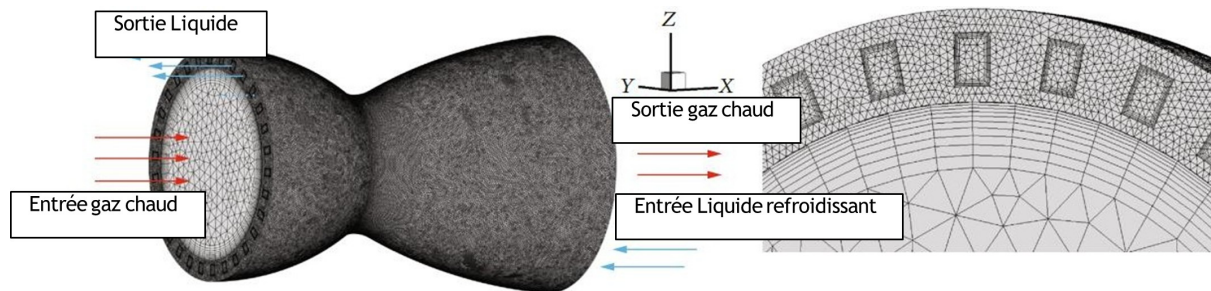


FIGURE 11 – Fluid + solid zone mesh

After simulation in ANSYS Fluent + ANSYS Thermal, the following results were obtained :

Axial position (mm)	Temp. (K)	γ	Mach	Viscosity (10^{-4} Pa·s)	C_p (J/mol·K)	Pr
0	3448.78	1.1251	0	1.1437	2278.4	0.5649
20.5	3442.70	1.1250	0.192	1.1422	2278.0	0.5651
47.3	3287.31	1.1213	1.000	1.1040	2267.5	0.5716
79.8	2783.91	1.1125	2.332	0.9785	2225.3	0.5940
111.64	2648.67	1.1122	2.635	0.9444	2210.7	0.6000

TABLE 2 – Thermal data of the gas in the combustion chamber at selected cross-sections

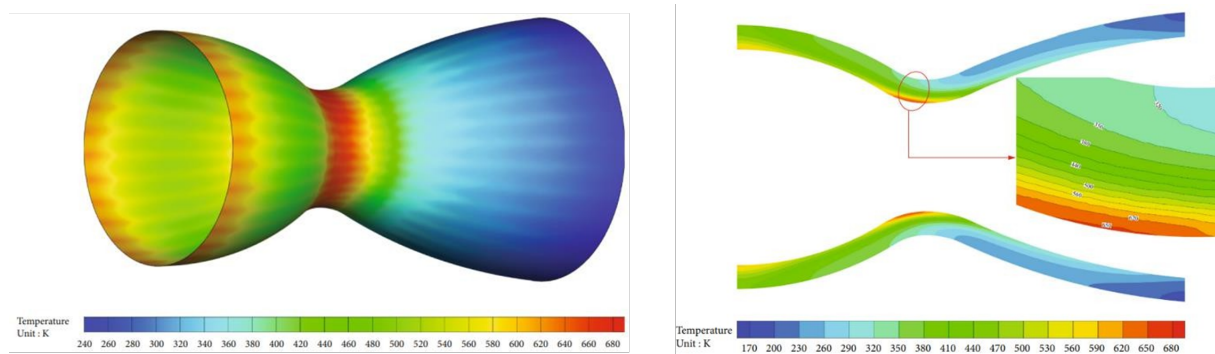


FIGURE 12 – Temperature distribution in the combustion chamber wall (cross-section taken between two channels)

The temperature peak is reached just before the narrowest cross-section, near the contraction zone of the chamber throat.

After varying parameters L and H in increments of 0.01 mm (between 1.5 and 5.0 mm for height and 0.8 and 3.0 mm for width), the following results were obtained :

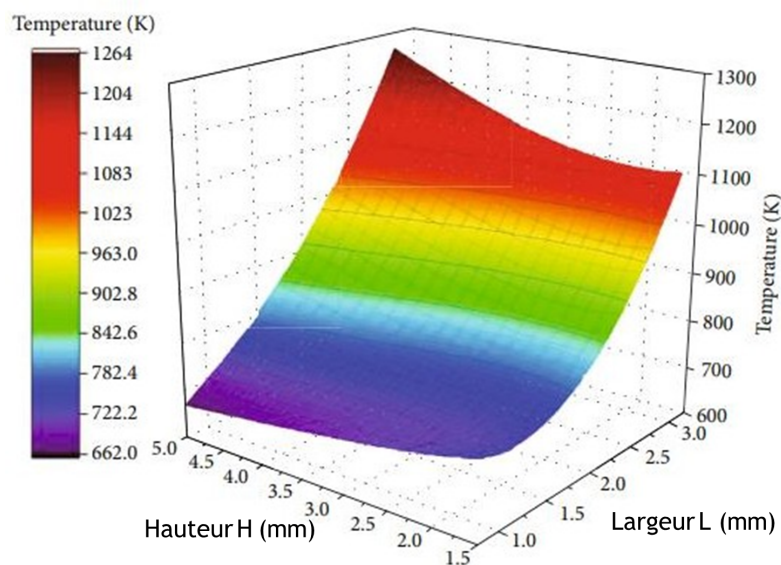


FIGURE 13 – Maximum combustion chamber temperature as a function of L and H

3.1.5 Conclusion

Optimal Result

The optimum is reached at $H = 5$ mm and $L = 0.8$ mm, with a minimum temperature of **586.6 K**.

Interpretation : When the channel width is increased, the coolant absorbs a large amount of heat at the inlet of the combustion chamber due to the large exchange surface area, in a region where its effect is not fully needed. By the time it reaches the chamber throat, where temperatures are highest, the coolant has already reached a significant temperature, reducing its ability to absorb heat efficiently.

Reducing the width while increasing the height allows the mass flow rate to be kept constant and delays heat absorption until the throat, where it is most needed.

3.2 Study 2 : Failure Prediction of the Starship Tank

Being a great fan of the **Starship** rocket project led by SpaceX, I have been closely following its development in detail since 2019. One of the major phases of this development is pressure testing of the LOX (liquid oxygen) and methane tanks.

The goal is to fill the tank with cryogenic liquid nitrogen and increase the internal pressure until rupture, in order to analyze the failure mode. Anticipating this test and having access to a tank drawing published online by an employee including dimensions and sheet thickness. I reconstructed the tank geometry and performed a hydrostatic pressure simulation to attempt to predict the failure zone and the critical pressure using the finite element method.

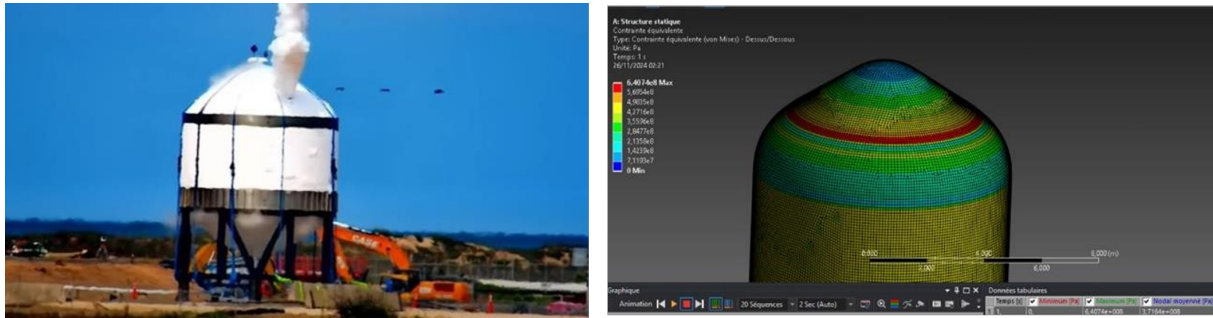


FIGURE 14 – FEA simulation and failure zone prediction

Results

- **Actual tank failure pressure** : 7.6 bar
- **Predicted failure pressure** : 8.1 bar
- **Difference** : Likely due to temperature being higher in the actual tests (the ultimate strength of stainless steel increases as temperature decreases)
- **Failure zone** : Predicted by the red zone (maximum stress). In the actual test, failure occurred exactly in the zone predicted by the finite element method : at the end of the arc and beginning of the linear section.

Watching the test live and accurately predicting the exact failure zone was particularly striking and validated the simulation approach used.

3.3 Study 3 : 6-DOF Rigid-Body Dynamics Simulator, RK4 Integrator

3.3.1 Objective

The goal is simple : obtain a numerically reliable integrator for the six degrees of freedom of a rigid body. No linearisation, no small-angle assumption, no analytical shortcut. Given a force and moment input at each instant, the integrator must propagate the full state of the body forward in time with controlled numerical error and remain stable over long simulation horizons.

Full implementation : github.com/floaty-bone/rocket-integrator

3.3.2 State Representation

The state of the rigid body is encoded in a 13-element vector :

$$\mathbf{s} = \left[\underbrace{x, y, z}_{\text{position}}, \underbrace{q_w, q_x, q_y, q_z}_{\text{attitude}}, \underbrace{v_x, v_y, v_z}_{\text{velocity}}, \underbrace{\omega_x, \omega_y, \omega_z}_{\text{angular velocity}} \right]$$

Attitude is represented by a unit quaternion rather than Euler angles. This eliminates gimbal lock entirely and keeps the kinematics well-conditioned at every orientation, including when the body undergoes large rotations.

3.3.3 Equations of Motion

The state derivative $\dot{\mathbf{s}}$ is assembled from four coupled equations evaluated at each integration step.

Position kinematics.

$$\dot{\mathbf{r}} = \mathbf{v}$$

Quaternion kinematics. The attitude evolves according to :

$$\dot{\mathbf{q}} = \frac{1}{2} \Omega(\boldsymbol{\omega}) \mathbf{q}$$

where $\Omega(\boldsymbol{\omega})$ is the 4×4 skew-symmetric matrix constructed from the body-frame angular velocity :

$$\Omega(\boldsymbol{\omega}) = \begin{pmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{pmatrix}$$

Translational dynamics. Newton's second law in the inertial frame :

$$\dot{\mathbf{v}} = \frac{1}{m} \mathbf{R}(\mathbf{q}) \mathbf{F}_{\text{body}} + \mathbf{g}$$

where $\mathbf{R}(\mathbf{q})$ is the rotation matrix from body frame to inertial frame, derived directly from the quaternion :

$$\mathbf{R}(\mathbf{q}) = \begin{pmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_w q_z) & 2(q_x q_z + q_w q_y) \\ 2(q_x q_y + q_w q_z) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z - q_w q_x) \\ 2(q_x q_z - q_w q_y) & 2(q_y q_z + q_w q_x) & 1 - 2(q_x^2 + q_y^2) \end{pmatrix}$$

Rotational dynamics. Euler's rotation equation in the body frame :

$$\dot{\boldsymbol{\omega}} = \mathbf{I}^{-1}(\mathbf{M} - \boldsymbol{\omega} \times (\mathbf{I} \boldsymbol{\omega}))$$

The term $\boldsymbol{\omega} \times (\mathbf{I} \boldsymbol{\omega})$ is the gyroscopic term. It accounts for the redistribution of angular momentum between axes whenever the angular velocity vector is not aligned with a principal axis of inertia. It is this term that governs all non-trivial rotational behaviour.

3.3.4 The RK4 Scheme

The four equations above are bundled into a single state-derivative function $f(\mathbf{s})$. The classical fourth-order Runge-Kutta method advances the state by one time-step h :

$$\begin{aligned}\mathbf{k}_1 &= f(\mathbf{s}_n) \\ \mathbf{k}_2 &= f\left(\mathbf{s}_n + \frac{h}{2} \mathbf{k}_1\right) \\ \mathbf{k}_3 &= f\left(\mathbf{s}_n + \frac{h}{2} \mathbf{k}_2\right) \\ \mathbf{k}_4 &= f(\mathbf{s}_n + h \mathbf{k}_3) \\ \mathbf{s}_{n+1} &= \mathbf{s}_n + \frac{h}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)\end{aligned}$$

The local truncation error is $\mathcal{O}(h^5)$, giving a global error of $\mathcal{O}(h^4)$. At $h = 1$ ms, this is more than sufficient for the dynamics of interest.

Numerical reliability. Two measures preserve long-term accuracy. The inverse inertia tensor $\mathbf{I}^{-1}$ is computed once at initialisation and cached, avoiding repeated matrix factorisation at every step. The quaternion is re-normalised after every step to prevent slow numerical drift from corrupting the attitude representation :

$$\mathbf{q} \leftarrow \frac{\mathbf{q}}{\|\mathbf{q}\|}$$

3.3.5 A Fun Observation : the Intermediate Axis Theorem

Once the integrator was running, the inertia tensor was set to three distinct principal moments, $I_1 = 300$, $I_2 = 100$, $I_3 = 30 \text{ kg} \cdot \text{m}^2$, and the body was given an initial angular velocity nearly aligned with the intermediate axis, with a small perturbation on the other two. The gyroscopic term in Euler's equation then drives the system into an unstable regime : the Dzhanibekov effect. The body undergoes periodic half-turns and re-aligns, indefinitely. The real-time 3-D animation made this directly visible, which is something no closed-form solution could have offered as clearly.

Key Outcomes

- Full 6-DOF rigid-body propagation with quaternion kinematics, with no gimbal lock.
- Fourth-order Runge-Kutta at $h = 1$ ms, global error $\mathcal{O}(h^4)$.
- Quaternion re-normalisation at every step to prevent long-term attitude drift.
- Numerical reproduction of the Dzhanibekov effect : rotation about the intermediate inertia axis is unstable and the simulator captures it exactly.